

# Standalone A-to-B Navigation with G1 EDU in Gazebo

## Standalone Point-to-Point Navigation Benchmark for Indoor Robot Operation

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# Research Goals

- Develop a reliable indoor A-to-B navigation Cognitive Architecture for the Unitree G1 EDU in Gazebo.
- Configure a complete ROS2 Nav2 pipeline
- Compare DWB, MPPI, and RPP local planners
- Test five indoor benchmark scenarios
- Evaluate success, time, path efficiency, collisions, recovery, and goal accuracy
- Identify the most suitable planner for the robot platform

## Main Challenge

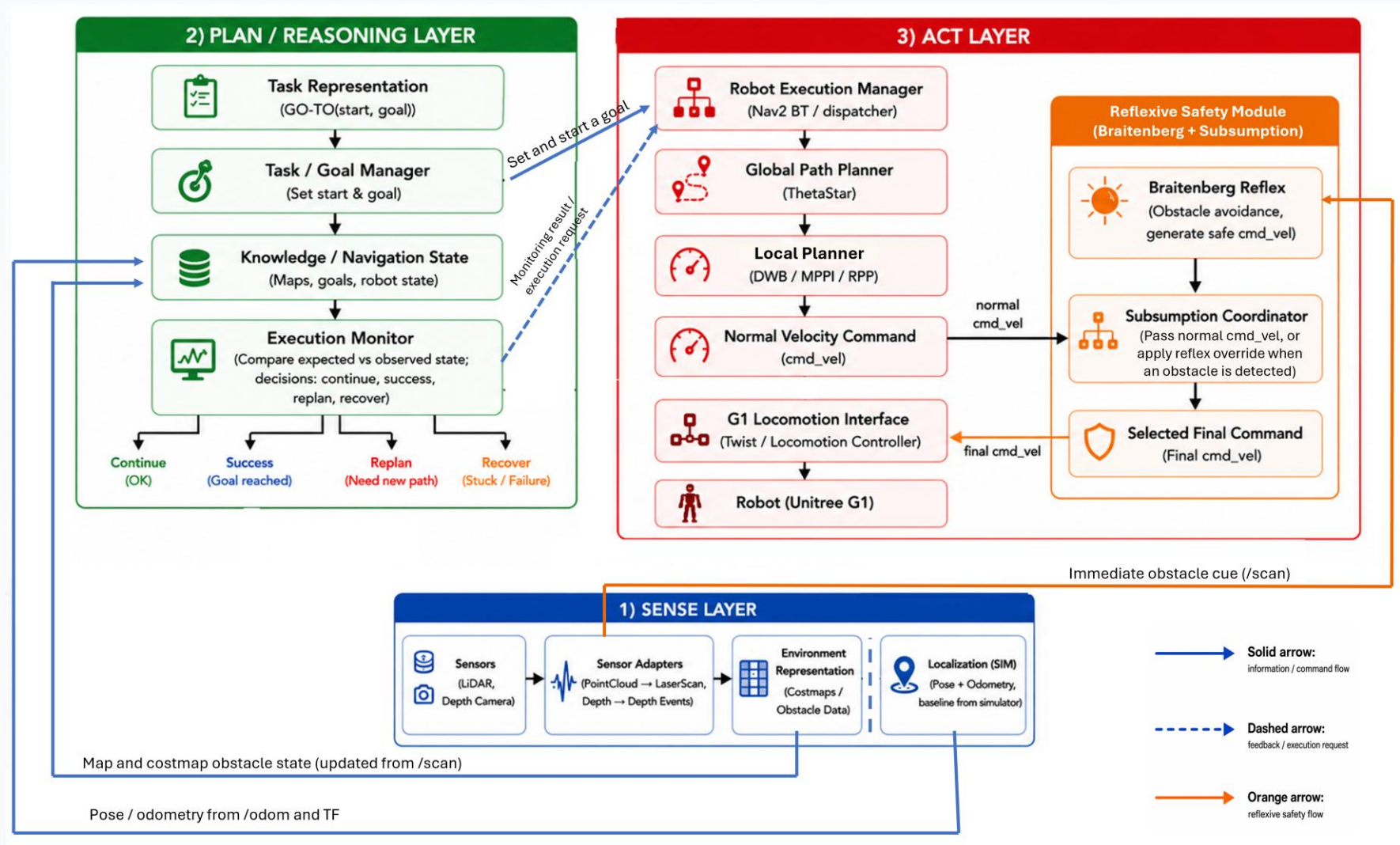
The Unitree G1 is not a small differential-drive robot in spirit: it has a larger footprint, a humanoid body with 29 DOF structure, and must navigate through constrained indoor layouts without collisions.

## Evaluation Setup

A complete simulation pipeline: known map, simulator localization, global path planning, local obstacle avoidance, behavior recovery, and run-to-run repeatability.



# Hybrid Reactive–Deliberative Cognitive Architecture



# Hybrid Reactive–Deliberative Cognitive Architecture

Hierarchical navigation with execution monitoring and fast bio-inspired safety

## Hierarchical Navigation — Deliberative Path

- The navigation objective is represented as a **GO-TO(start, goal)** task .
- The Robot Execution Manager coordinates **ThetaStar global planning** and the selected local planner: **DWB, MPPI, or RPP**.
- Execution is continuously monitored using the observed pose, progress, clearance, controller status, and goal state.
- Based on the monitoring result, the system may **continue, replan, recover, succeed, or fail**.

## Bio-inspired Reflexive Safety — Reactive Path

- Immediate obstacle information activates a **Braitenberg-inspired reflex**.
- The reflex generates a short-term safe velocity command without waiting for full replanning.
- Through priority arbitration, the safety command can temporarily override the normal velocity command.

The architecture combines a hierarchical deliberative navigation path with a fast reflexive safety pathway.

# ROS 2 implementation mapping

How the cognitive architecture maps to concrete topics, actions, services, and TF

## Sense and observed state

- The 3D LiDAR data are adapted into a two-dimensional `/scan` stream for navigation and obstacle detection.
- The Global Costmap combines the static map with obstacle updates, while the Local Costmap represents nearby obstacles for local motion control.
- `/odom` and the TF chain provide the robot pose and observed motion state.
- The Benchmark Logger records `/odom`, `/cmd_vel`, `/scan`, and `/reflex_active` to compute trajectory, clearance, motion, and reflex metrics.

## Planning, control and safety

- Navigation goals are sent through the long-running Nav2 `/navigate_to_pose` action.
- The Global Planner uses ThetaStar to generate the route, while the selected Local Planner—DWB, MPPI, or RPP—produces the motion command.
- The Local Planner publishes the normal command on `/nav_cmd_vel`. The subsumption coordinator forwards it during normal operation or selects `/reflex_cmd_vel` when the reflex is active. The final command is published on `/cmd_vel`.
- Services handle discrete operations such as enabling the logger and spawning or deleting the dynamic obstacle.

## ROS communication roles



### Topics

Continuous sensor, state, and command streams



### Action

`/navigate_to_pose`



### Services

discrete operations



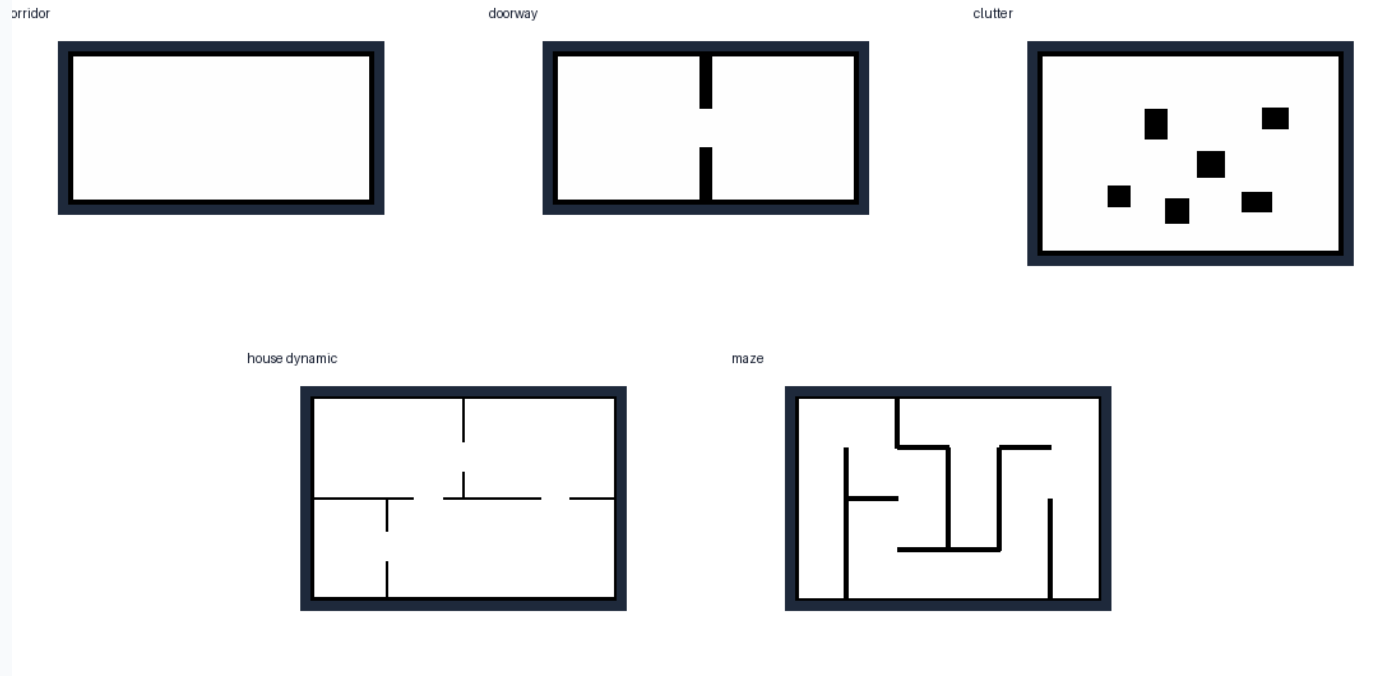
### TF

`map → odom → base_link`

**Key idea: ROS 2 keeps the cognitive layers implementable through topic streams, action feedback, service calls, and TF frames.**

# Benchmark scenarios and metrics

Five indoor tasks stress different navigation behaviors.



## Scenario roles

corridor = baseline timing; doorway = narrow passage; clutter = local obstacle adaptation; house\_dynamic = moving blockage; maze = high-curvature route.

## Metrics

success rate, completion time, collision count, recovery count, reflex count, final goal error.

## Run matrix

5 scenarios  $\times$  3 planners  $\times$  3 repetitions = 45 runs. Each run writes telemetry, per-run summary CSV, aggregate CSV and plots.

# Benchmark scenarios and metrics

## Corridor Scenario

**DWB**

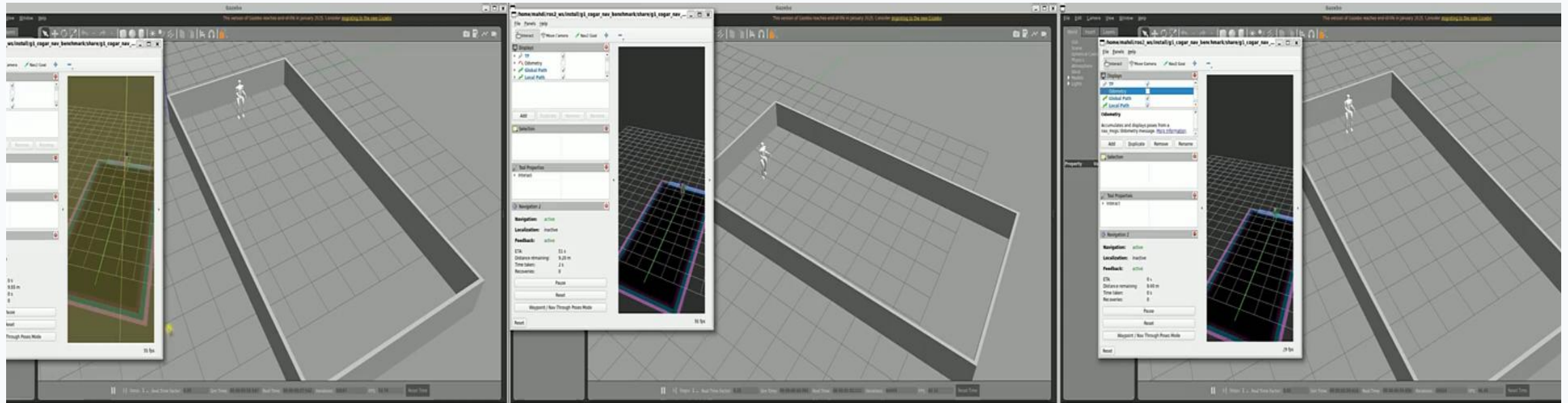
planner

**MPPI**

planner

**RPP**

planner



# Benchmark scenarios and metrics

## Doorway Scenario

**DWB**

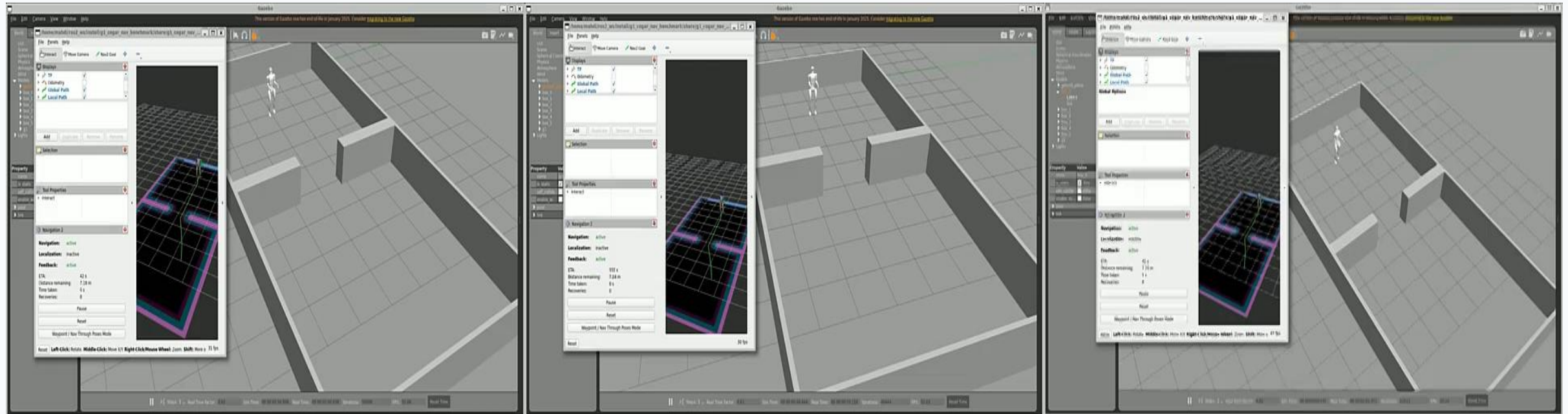
planner

**MPPI**

planner

**RPP**

planner



# Benchmark scenarios and metrics

## Clutter Scenario

**DWB**

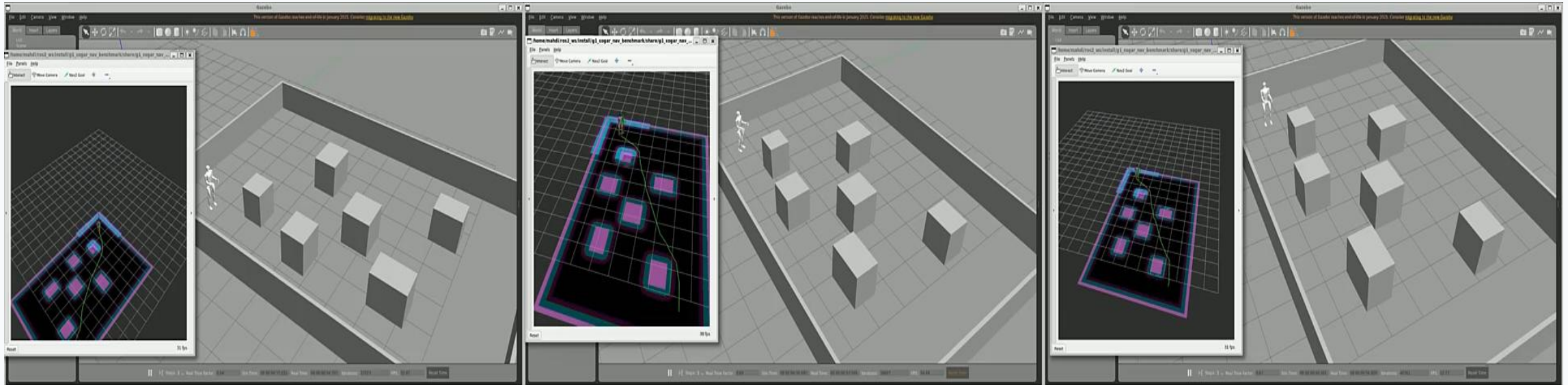
planner

**MPPI**

planner

**RPP**

planner



# Benchmark scenarios and metrics

## Dynamic Blockage Scenario

**DWB**

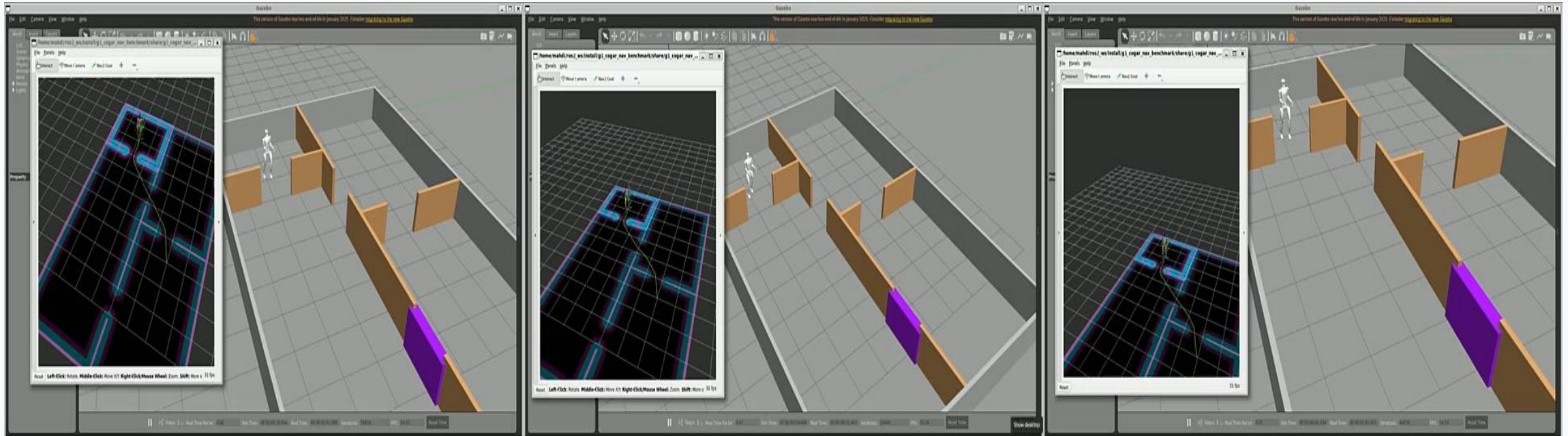
planner

**MPPI**

planner

**RPP**

planner



# Benchmark scenarios and metrics

## Maze scenario

**DWB**

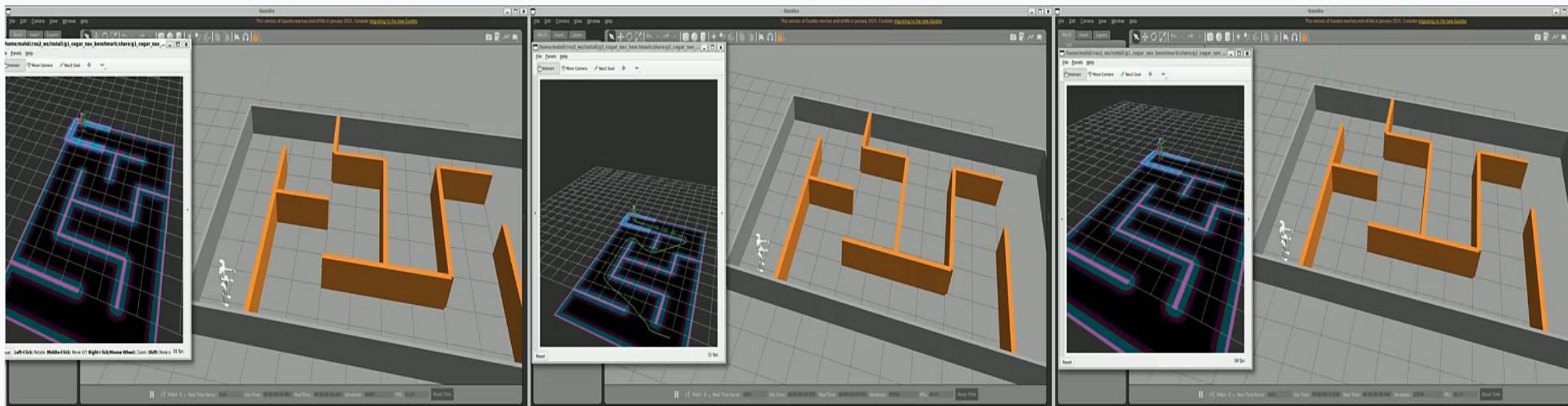
planner

**MPPI**

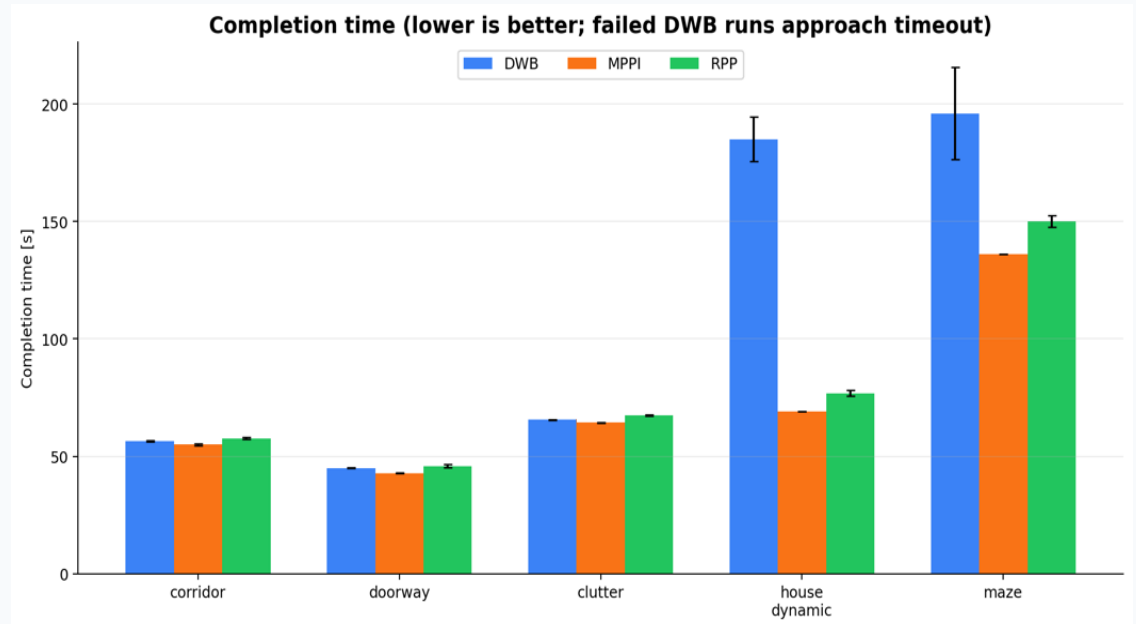
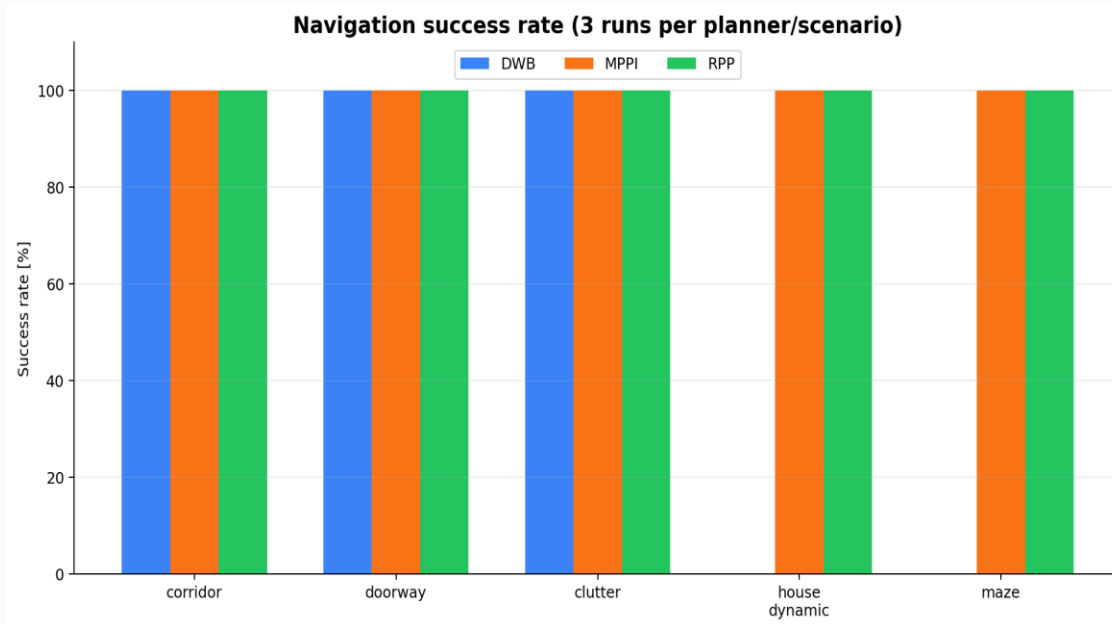
planner

**RPP**

planner



# Results I: success and completion time



## Main observation

DWB reached all simple tasks but failed in house\_dynamic and maze. MPPI and RPP completed all tasks.

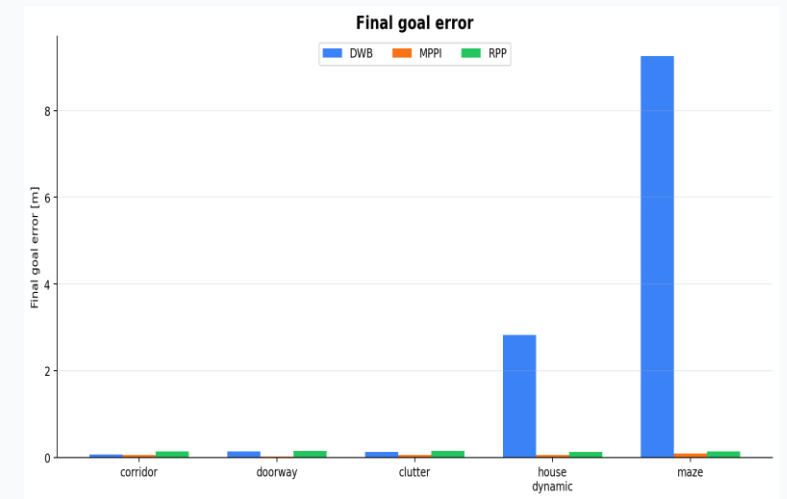
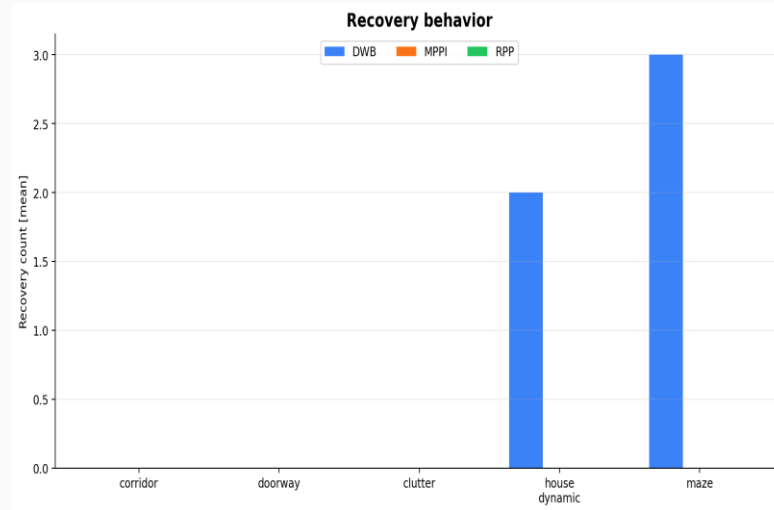
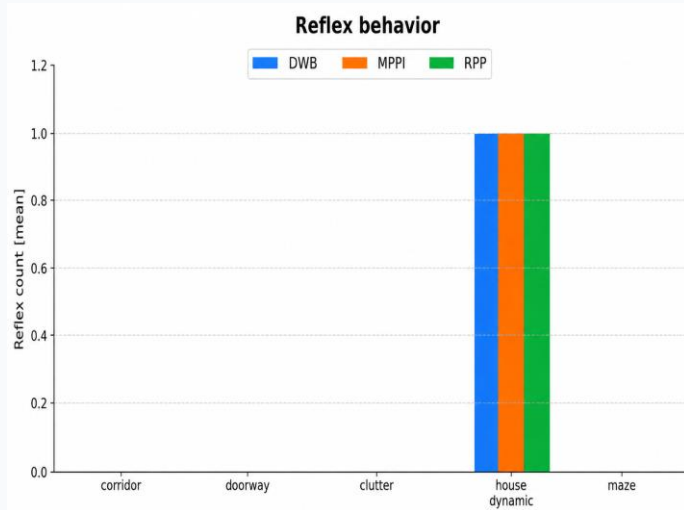
## Timing

MPPI was generally the fastest among the successful planners in the complex scenarios.

## Interpretation

The results match the planner properties: predictive MPPI handles complex local decisions better than other local planners.

# Results II: recovery, goal error and reflex



## Goal error

MPPI showed the smallest final goal error in most successful tasks. RPP often stopped safely but less precisely.

## Recovery

DWB triggered recovery in the difficult scenarios. MPPI/RPP usually completed the difficult scenarios through normal navigation and replanning, without invoking Nav2 recovery.

## Reflex

Braitenberg briefly overrode `cmd_vel` in `house_dynamic` to avoid the sudden obstacle.

## Safety

No collision-proxy events were recorded

# Planner-by-planner interpretation

| Planner | Success  | Mean time | Mean recovery | Mean final error |
|---------|----------|-----------|---------------|------------------|
| DWB     | 60% avg  | 109.3s    | 1.0           | 2.48m            |
| MPPI    | 100% avg | 73.5s     | 0.0           | 0.05m            |
| RPP     | 100% avg | 79.6s     | 0.0           | 0.14m            |

## DWB: baseline

Good in corridor, doorway and clutter.  
Weak in maze and dynamic-obstacle conditions under this configuration. The failures are visible in success rate, recovery and final error.

## MPPI: best overall

The most consistent profile in this benchmark: 100% success, strong completion time, low final error, and no collisions.

## RPP: robust but conservative

Reached the goal in all scenarios, but generally required more time and had slightly larger final error than MPPI.

# Congruence of results and conclusions

## Supported conclusion 1

In the tested Gazebo scenarios, MPPI is the best overall planner profile for this G1 EDU navigation pipeline.

## Supported conclusion 2

RPP is robust and conservative, but its regulated behavior makes it slower and slightly less precise at the final pose.

## Supported conclusion 3

DWB is suitable as a simple baseline, but not robust enough for the hardest maze and dynamic cases with my tuning.

## Limitations

Simulation-only evaluation, only three repetitions, and simplified humanoid locomotion without full-body dynamics.

## TEB note

TEB was suggested in the assignment as an example, but I used RPP as the third planner because a stable ROS2 Humble TEB profile was not available in my environment.

## What I would improve

Increase the number of experimental repetitions, test the system on the real G1 and in dynamic-human scenarios, and perform parameter sensitivity analysis. With stronger computing hardware, the benchmark could also be extended to full-body dynamic simulation with balance control, gait generation, and more realistic humanoid locomotion for the 29-DOF robot.

# Final takeaway

**The implemented system integrates hierarchical navigation, execution monitoring, reflexive safety, and quantitative evaluation.**

**ROS2**

modular package

**Nav2**

planner profiles

**Gazebo**

benchmark worlds

## Conclusion:

**MPPI is the recommended local planner for the tested G1 Gazebo benchmark. RPP is a robust but more conservative alternative, while DWB remains a useful baseline that reveals limitations in difficult, high-curvature scenarios and recovery behavior.**